5 A double slit consists of two parallel slits, each of width 0.12 mm. The separation of the slits is 2.8 mm, as illustrated in Fig. 5.1.

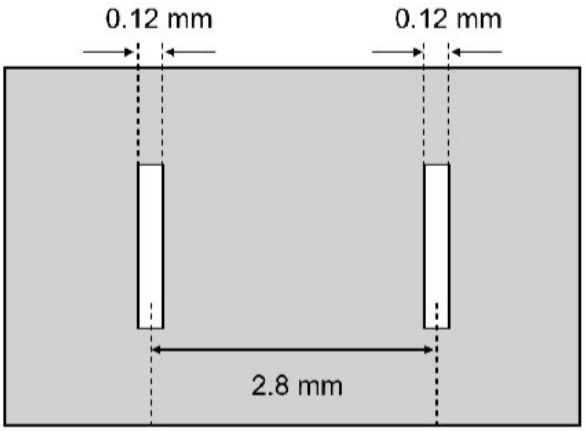



**Fig. 5.1** (not to scale)

Parallel light of wavelength 670 nm is incident normally on the double slit.

A screen is placed parallel to the plane of the double slit at a distance of 1.2 m from the slits, as illustrated in Fig. 5.2

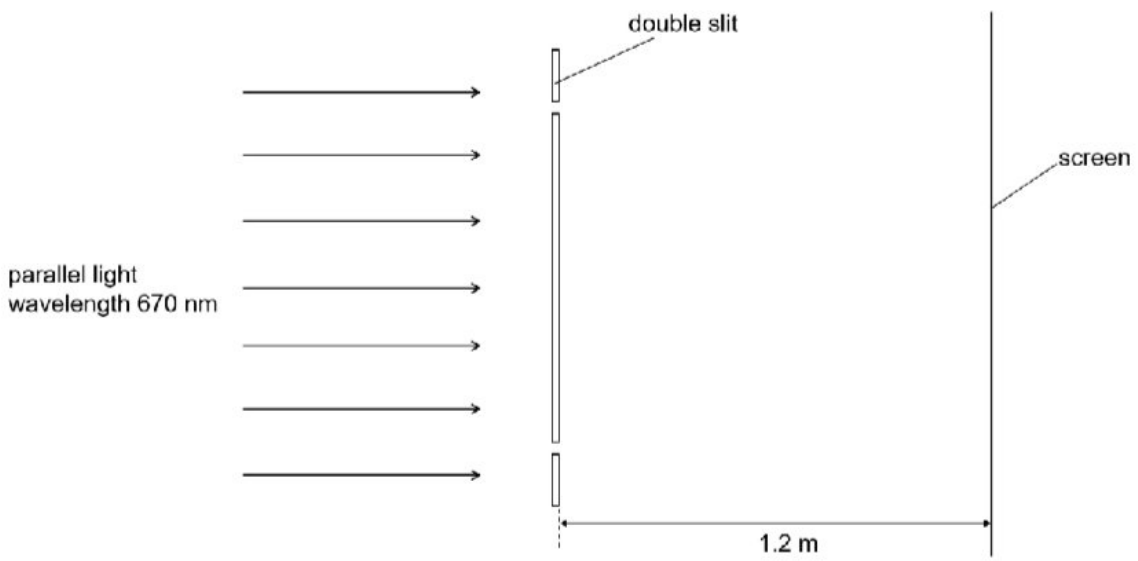



**Fig. 5.2** (not to scale)

**(a)** Initially, one of the two slits is covered.

Show that the width of the central fringe for a single-slit diffraction pattern on the screen is 13.4 mm.

[3]

**(b)** Both slits are now uncovered.

**(i)** Use the Rayleigh criterion to explain whether the diffraction patterns produced by the two slits are seen on the screen as being separated.

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.....

.....

..... [2]

**(ii)** Using your answer in **(a)**, estimate the number of fringes within the central bright maximum.

number = ..... [2]

[Total: 7]